Aryan Jalota

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Education

The University of Texas at Austin — Austin, TX
Bachelor of Science in Computer Science & Mathematics, Minor in Statistics — *Aug 2024 – May 2028*

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming

Experience

Salesforce — May 2026 – Aug 2026
Applied AI Software Engineering Intern — *San Francisco, CA*

- Architected and shipped a distributed production platform for Marc Benioff and Salesforce's executive leadership team; built a shared Node.js/TypeScript backend with 40+ REST APIs and an MCP tool server supporting enterprise workflows.
- Improved AI query accuracy from 21% to 95% and reduced p50 latency 5× through deterministic entity resolution, LLM routing, and multi-layer caching across live enterprise data.
- Reduced executive briefing preparation time 92%, from 2 hours to 10 minutes, by building production retrieval and automation workflows across enterprise systems.
- Improved deployment reliability and scalability by migrating a production backend from SQLite to PostgreSQL across 20+ tables and implementing OAuth 2.0/Google SSO.

Depository Trust & Clearing Corporation — June 2025 – Aug 2025
Software Engineering Intern — *Coppell, TX*

- Built scalable Python pipelines to automate audit-report extraction and validation, processing 1,000+ pages/hour with structured logging, fault tolerance, and automated validation, reducing workload by 10–15 hours/week.
- Built a developer productivity tool that instrumented user workflows and automatically generated process documentation, reducing internal documentation time 50–60%.

GISense Lab – The University of Texas at Austin — Aug 2025 – May 2026
Undergraduate Research Software Engineer – Multimodal AI — *Austin, TX*

- Improved ML training throughput 45% by designing a distributed data pipeline using parallel and asynchronous processing to transform geospatial datasets into optimized HDF5 feature stores.
- Built modular end-to-end ML pipelines in Python using PyTorch and NumPy with PyTest, CI workflows, and automated data validation, improving reliability and reproducibility across data-intensive research systems.

Traffic Monitoring Lab – The University of Texas at Austin — Aug 2024 – May 2026
Undergraduate Research Software Engineer – Computer Vision — *Austin, TX*

- Designed and evaluated real-time computer vision systems using PyTorch, YOLOv8, and OpenCV with GPU-accelerated training and inference for low-latency vehicle detection.
- Achieved 0.85+ mAP with high-80s precision/recall through systematic model training and evaluation workflows for deployment-oriented traffic analysis.

Projects

Distributed Lakehouse Analytics Platform | *Python, Apache Spark, Delta Lake, FastAPI, Redis*

- Architected a distributed lakehouse using Apache Spark and Delta Lake, implementing ACID transactions and SQL analytics across 500M+ records.
- Reduced repeated-query latency 60% with a FastAPI query service using Redis caching and asynchronous job scheduling.

Distributed Search Engine | *C++, Go, gRPC, Redis, RocksDB, Docker*

- Built a distributed search engine indexing 10M+ documents with inverted indexes and BM25 ranking at sub-100 ms query latency across shards.
- Reduced query latency 65% with a distributed gRPC query engine using consistent hashing and Redis caching; parallelized RocksDB indexing increased throughput 4×.

High-Performance Trading Engine | *C++17, Multithreading, Linux, CMake, Google Benchmark*

- Engineered a cache-efficient C++17 limit order book processing 1M+ market events/sec using custom memory management and performance benchmarking.
- Achieved 3× higher throughput by parallelizing order processing with std::thread and producer–consumer queues.

Technical Skills

Languages: C++17, Python, Go, Java, TypeScript, JavaScript, SQL
Systems: Distributed Systems, Concurrency, Parallel Computing, Operating Systems, Microservices, gRPC, REST APIs
Data: Apache Spark, Delta Lake, PostgreSQL, MySQL, SQLite, Redis, RocksDB, HDF5
Infrastructure: Linux, Docker, Kubernetes, Git, CI/CD, AWS
Frameworks: Node.js, FastAPI, React, Spring Boot, PyTorch